

GARRACH MALEK

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Grand Rapids, MI 49534

PROFILE

Graduate student at the College of Computing, GVSU, and recent engineering graduate from Sup'Com with a Master's degree in Management of Technology and Innovation. Creative, adaptable, and solution-oriented, with a strong passion for driving innovation. Skilled in programming, optimization, and problem-solving, with a proven ability to thrive in collaborative environments and adapt quickly to new challenges.

EXPERIENCE

- **Research Graduate Assistant at College of Computing - GVSU**  August – Present, 2025
Computer Vision Developer Grand Rapids, United States of America
 - Designed and implemented an end-to-end continuous learning architecture, including automated teacher model retraining, student model distillation, and deployment to Raspberry Pi devices for real-time plant health monitoring with robust and scalable computer vision pipelines.
- **Intern at College of Computing - GVSU**  March – July, 2025
Computer Vision Developer Grand Rapids, United States of America
 - Developed a computer vision classification model to assess the health status of lettuce plants, effectively distinguishing between healthy and diseased samples. Applied transfer learning and model distillation techniques to enhance performance with limited data.
- **Intern at EY (Ernst & Young)**  September - December, 2024
Generative AI Intern Tunis, Tunisia
 - Developing a solution that leverages generative AI to automate and optimize code testing. The project aims to enhance the accuracy of testing processes, identify anomalies, and predict potential errors.

EDUCATION

- **Grand Valley State University - College of Computing** 2025 – 2026
Master of Science in Applied Computer Science Grand Rapids, MI, USA
- **Higher School of Communications of Tunis (Sup'Com)** 2022 – 2025
Master of Science in ICT Engineering - Specialization: Management of Technology and Innovation Ariana, Tunisia
- **Preparatory Institute for Engineering Studies of Nabeul (IPEIN)** 2020 – 2022
Selective Two-Year Engineering Preparatory Program - Ranked 27/830 in Engineering Track Nabeul, Tunisia

PROJECTS

- **Semantic segmentation using deep learning for wildlife detection** February 2024 - May 2024 
Tools: Pytorch, TensorFlow, CVAT.ai
 - Led data preparation for a wildlife dataset, including generating initial masks using SAM and annotating 10,000 images using the CVAT.ai platform.
 - Fine-tuned three models—SAM, Segformer, and UNet—evaluating and comparing their performance to determine the most effective solution for the dataset.

VOLUNTEERING EXPERIENCE

- **IEEE/ACM MODELS 2025** 5–10 October 2025
Student Volunteer Grand Rapids, United States of America
 - Hosted several online Zoom sessions and assisted on-site with registration and attendee coordination during the 28th IEEE/ACM International Conference on Model-Driven Engineering Languages and Systems (MODELS 2025).

SKILLS

- **Soft skills:** Problem-solving, Critical thinking
- **Programming Languages:** Python, C++
- **Web Technologies:** ReactJS, Django, MySQL, PostgreSQL
- **Data Science & Machine Learning:** Transformers, Retrieval-Augmented Generation (RAG), Fine-Tuning, Multi-Agent Systems, LangChain, LangGraph

ADDITIONAL INFORMATION

Languages: English (B2), French (B2), Arabic (native language)

Interests: Chess, Problem Solving, Football, Music